

COLLEGE ERP SYSTEM

SYSTEM DEPLOYMENT & USER MANUAL

Console Layered MVC Architecture - Java & MySQL Platform

Version: 1.0.0 (Release-Ready)

Target Audience: System Administrators, Operators, and Testers

Authoring: Google DeepMind team pair programming agent

TABLE OF CONTENTS

1. System Prerequisites & Hardware Specs	Page 2
2. Database Operations & Table Import Instructions	Page 3
3. Compilation & Launching Command Lines	Page 4
4. Navigation Guide - Input Safety & Tabular Results	Page 5
5. System Troubleshooting & Conflict Resolution	Page 6

JULY 2026

1. System Prerequisites & Hardware Specs

This section outlines the requirements and credentials configuration to launch the application.

1.1 Runtime Requirements

- Java SE Runtime Environment: JDK version 22.0.2 or higher is required.
- Database Engine: MySQL Community Server version 8.0.x or higher.
- JDBC Driver: mysql-connector-j-9.0.0.jar (installed inside the /lib folder).
- Terminal compatibility: Powershell, cmd, or git bash with ANSI color coding enabled.

1.2 Database Credentials Configuration

Database connection strings are defined in the AppConstants.java utility file. Ensure values correspond ex

```
// Located in: src/constants/AppConstants.java
public static final String DB_URL = "jdbc:mysql://localhost:3306/college_erp";
public static final String DB_USERNAME = "root";
public static final String DB_PASSWORD = "jaideep@2006";
```

1.3 Test Database Overrides

To support automated testing without polluting production tables, the connection engine implements syste

2. Database Schema Configuration

Follow these commands to import the database structures and sample datasets into MySQL.

2.1 Schema Initialization

Log in to your local MySQL instance and run the schema queries to construct the relational table tables, ke

```
# PowerShell commands to initialize the databases
# 1. Create and populate the primary production database
& "C:\Program Files\MySQL\MySQL Server 8.0\bin\mysql.exe" -u root -pjaideep@2006 < database/schema.sql
& "C:\Program Files\MySQL\MySQL Server 8.0\bin\mysql.exe" -u root -pjaideep@2006 < database/insert_sampl
```

2.2 Test Database Setup

To run tests in isolation, set up the college_erp_test instance to support AppTest validation. This ensures pr

```
# 2. Recreate schema and mock data for the test database Target
& "C:\Program Files\MySQL\MySQL Server 8.0\bin\mysql.exe" -u root -p -e "CREATE DATABASE college_erp_test"
(Get-Content database/schema.sql) -replace 'college_erp', 'college_erp_test' | Set-Content database/sche
Get-Content database/schema_test.sql | & "C:\Program Files\MySQL\MySQL Server 8.0\bin\mysql.exe" -u root
```

2.3 Table Dependencies

Keep in mind that foreign key constraints exist on tables: subjects, attendance, marks, fees, and book_issu

3. Compilation & Launching Command Lines

Execute these terminal commands to compile compile-safe artifacts and start the application.

3.1 Compile Instructions

Run this compilation command from the root folder. It compiles source files into the /bin folder, utilizing the

```
# Compile all packages inside the project
javac -d bin -cp "lib/mysql-connector-j-9.0.0.jar" src/config/*.java src/constants/*.java src/enums/*.ja
```

3.2 Execute Application

Start the interactive console application by loading the Main execution class.

```
# Start the interactive College ERP Application CLI
java -cp "bin;lib/mysql-connector-j-9.0.0.jar" app.Main
```

3.3 Run Automated Integration Tests

Run the self-contained test runner suite against the test target to verify CRUD operations.

```
# Run the complete integration test runner against college_erp_test
java -cp "bin;lib/mysql-connector-j-9.0.0.jar" app.AppTest
```

4. Navigation & Troubleshooting

4.1 Input Safety and Validation Utilities

The console interface is equipped with robust parser validators to avoid application termination:

- Integers, doubles, and dates are caught inside a loop via InputUtil.
- Name fields, phone formats, and email patterns are validated before SQL executions.
- Output prints use aligned column widths via TablePrinter.

4.2 Default Administrator Login Credentials

To enter the administrative panel, utilize the mock credential entries initialized in the insert script:

```
Username : admin  
Password : admin123
```

4.3 Troubleshooting Common Conflicts

- Error: 'Access denied for user': Check DBConnection.java password variables match MySQL Server details
- Error: 'Class Not Found Exception': Ensure you include the mysql JAR inside the classpath during compilation
- Error: 'Cannot delete/update parent row': Check references; students, faculty, or subjects may have references
- Warning: 'ANSI escape codes printed as text': Ensure your terminal client supports color sequences (PowerShell)